

§1.564 x is a cover $\Rightarrow R$ entire — `tabulated_is_entire_iff_left_cover`

Panel 0 — pullback of (y,y) : pair $\langle 1_T, 1_T \rangle$ over y induces unique lift $d: T \rightarrow P$ with $d \gg l = 1$, $d \gg r = 1$.

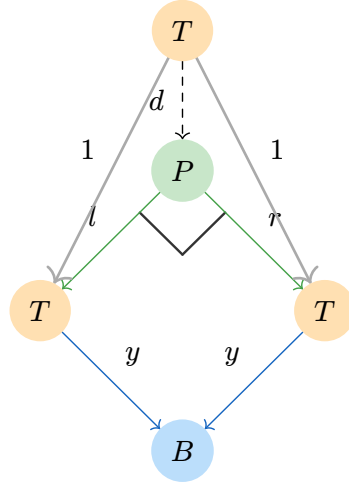


Figure 1: $hd_1: d \gg l = 1_T$, $hd_2: d \gg r = 1_T$, pullback condition: $l \gg y = r \gg y$

Panel 1 — composition RR° : pullback of (y,y) , image of $\langle l \gg x, r \gg x \rangle$ into $A \times A$. $RR^\circ = (I; i \gg fst, i \gg snd)$.

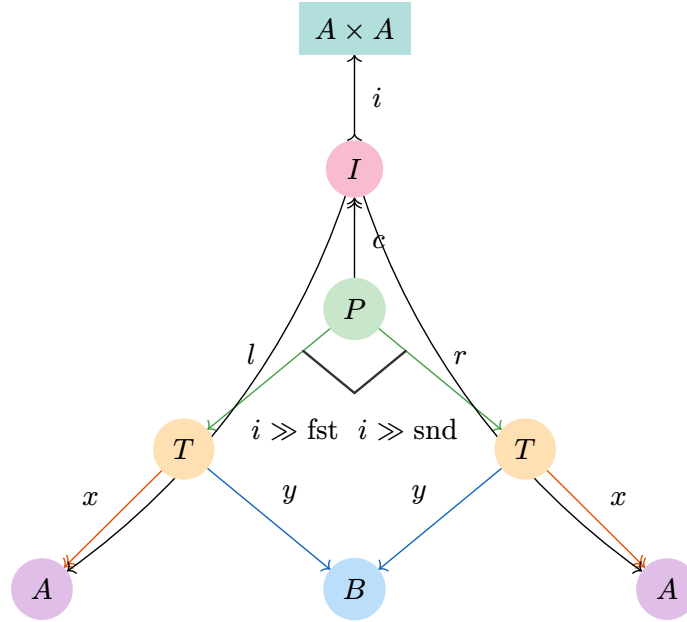


Figure 2:

Panel 2 — $J :=$ pullback of (Δ, i) . t lifts $\langle x, d \gg c \rangle$. $x = t \gg k$ cover, k monic $\Rightarrow k$ iso (§1.363). $k^{-1} \gg j: A \rightarrow I$ witnesses $1 \sqsubseteq RR^\circ$.

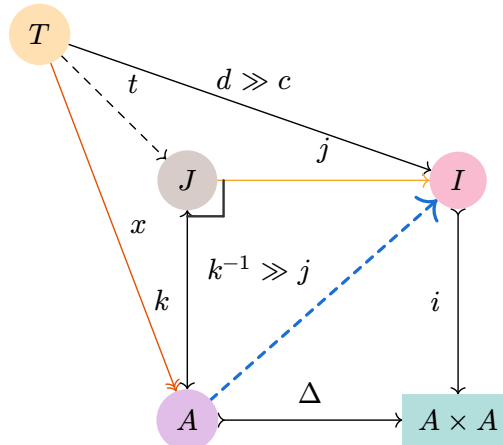


Figure 3: $hdl: (d \gg c) \gg i = x \gg \Delta$, $t := pbJ.lift \langle T, x, d \gg c \rangle$, $ht: t \gg k = x$. k monic, $x = \text{cover} \Rightarrow k$ iso. $h := k^{-1} \gg j$ satisfies $pf_1: h \gg (i \gg fst) = 1$, $pf_2: h \gg (i \gg snd) = 1$.

Legend: $\rightarrow$ hollow head = cover, $\rightarrow$ tailed = monic, double arrow = iso, dashed = induced / witness, blue = witness, $\ulcorner$ = pullback corner.